Sujit Debnath

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SUMMARY

An enthusiastic Computer Science graduate with proven experience in teaching, analysis, and volunteering who likes to take initiatives and seeks new challenges. I have a great interest in AI and Data Science, particularly devising a better problem-solving method for complicated tasks while learning new technologies.

EDUCATION

North South University — **Bashundhara R/A, Dhaka**
Bachelor of Science in Computer Science and Engineering; CGPA: 3.66 out of 4.00 — *2015 – 2018*

Dhaka College — **Dhanmondi, Dhaka**
Higher Secondary Certificate; Science Group; GPA: 5.00 out of 5.00 — *2011 – 2013*

Rampura Ekramunnesa Boys' High School — **Rampura, Dhaka**
Secondary School Certificate; Science Group; GPA: 5.00 out of 5.00 — *2009 – 2011*

EXPERIENCE

Project Coordinator — **Feb 2020 – Dec 2020**
***Robi Axiata Limited**, Department of Analytics Center of Excellence* — *Gulshan, Dhaka*

- Worked on a project whose goal is to build a custom conversational chatbot that can interact with group members of *Microsoft Kaizala*, follow instructions, and send feedback accordingly.
- **Responsibilities:**
  - Analyze network performance KPI of 2G, 3G, and LTE for both Ericsson and Huawei vendors.
  - Develop and tune SQL queries, including SQL based DB reporting.
  - Analyze and performance reporting (e.g., Pre-Post analysis, identifying bad performing cells, worse cells tracking, etc.) of ASC 10 to 12 MHz Expansion and U900 to L900 (3G to 4G) Conversion projects.

Lab Officer — **Jan 2019 – Dec 2019**
***North South University**, Department of Electrical & Computer Engineering* — *Bashundhara R/A, Dhaka*

- **Conduct lab classes for the following courses:**
  - CSE 115L (Programming Language I Lab): Summer 2019, Fall 2019
  - CSE 215L (Programming Language II Lab): Spring 2019, Summer 2019, Fall 2019
  - CSE 225L (Data Structures and Algorithms Lab): Fall 2019
  - CSE 311L (Database Systems Lab): Spring 2019
- **Responsibilities:**
  - Designing course curriculum and conducting lab classes.
  - Arrange and evaluate assignments, exams, and projects.
  - Invigilate during exams and assist for grading.

Under-Graduate Assistant — **Jan 2019 – Apr 2019**
***North South University**, Department of Electrical & Computer Engineering* — *Bashundhara R/A, Dhaka*

- **Assist Faculty members for the following course:** CSE 115 (Programming Language I), Instructors- *Dr. A. K. M. Ashikur Rahman* & *Dr. Muhammad Asif Hossain Khan*
- **Responsibilities:**
  - Conduct tutorial sessions for students needing extra help outside of class hours.
  - Grading of home-works, assignments and assist faculty members in their course-related work.

PUBLICATIONS

- Shakil Ahmed Sumon, Joydip Chowdhury, **Sujit Debnath**, Nabeel Mohammed, Sifat Momen, **"Bangla Short Speech Commands Recognition Using Convolutional Neural Networks"**, In *1st International Conference on Bangla Speech and Language Processing* (*ICBSLP*), pp. 1-6, 2018.
- Nazmul Alam Diptu, Md. Asif Khan, **Sujit Debnath**, Abdullah Al Imam, Al Mahadi Hasan Rakib, Kazi Asfaq Ahmed Ador, Rashedur M. Rahman, **"Early Detection of Glaucoma Using Fuzzy Logic in Bangladesh Context"**, In *9th International Conference on Intelligent Systems* (*IS'18*), pp. 87-93, 2018.

PROJECTS

- **Kothamala - A Bangla Speech Recognition Engine:** This research-based project was conducted in two divisions; one was Bengali Short Speech Commands Recognition, and the other one was Bengali Transcription. In the first division, a robust dataset had been collected with 50 classes, and three different Convolutional Neural Network (CNN) architectures had been proposed. In contrast, combined with a language model, another three different model architectures with Connectionist Temporal Classification (CTC) loss function had been experimented for building speech to text engine, where an open-source dataset was taken from *OpenSLR*, released by Google.
- **Early Risk Prediction of Stroke:** In this research-based project, several Machine Learning techniques such as Logistic Regression, Support Vector Classifier, etc along with Synthetic Minority Over-sampling Technique (SMOTE) had been used to predict the risk of stroke on 16 different features excluding the class.

SKILLS

- **Programming Languages:** Python (Frequently used), JAVA, C, C++, SQL
- **Machine Learning, Deep Learning and Visualization Libraries:**
  - Python based libraries: Numpy, Pandas, SciPy
  - Machine Learning library: Scikit-learn
  - Deep Learning library: Keras
  - Visualization: Matplotlib, Plotly, Seaborn
- **Database:** Relational database - Oracle, MySQL, and NoSQL database - MongoDB
- **Version Control System:** GitHub
- **Simulation Tools and Others:** Matlab, Logisim
- **Packages:** MS Word, MS PowerPoint, MS Excel, and LaTeX
- **Languages:** Bengali (Native), and English (Proficient)

EXTRACURRICULAR ACTIVITIES

NSU Problem Solvers (NSUPS) **Bashundhara R/A, Dhaka**
Member *2015 – 2016*

- NSUPS is a problem-solving community based on North South University and this community's primary purpose is to help beginner students get involved in competitive programming. I got involved with this community and had the opportunity to participate in several programming competitions organized by NSUPS.

JAAGO Foundation **Banani, Dhaka**
Volunteer *2014 – 2015*

- JAAGO Foundation works for the betterment of the nation through fulfilling the educational needs of underprivileged children. I attended several events organized by JAAGO, including World Environment Day.

Volunteer for Bangladesh (VBD) **Banani, Dhaka**
Community Volunteer *2014 – 2015*

- VBD is a non-political volunteer and youth wing of the JAAGO Foundation, which provides the youth of Bangladesh a voice and platform to contribute to nation-building. However, I attended several events organized by VBD, and some notable events were Universal Children's Day and The Great Kindness Challenge.

SCHOLARSHIP & ACHIEVEMENTS

- **Financial Aid:** Achieved 25% tuition waiver from North South University. *Summer 2015 – Fall 2018*
- **Honors in BS:** *magna cum laude*, Graduating class of 2019, 22nd Convocation of NSU. *March 2019*

REFERENCES

- **Dr. Mohammad Rashedur Rahman**
  Professor, Department of Electrical and Computer Engineering, North South University.
  rashedur.rahman@northsouth.edu ece.northsouth.edu/people/rashedur-rahman
- **Dr. Nabeel Mohammed**
  Associate Professor, Department of Electrical and Computer Engineering, North South University.
  nabeel.mohammed@northsouth.edu ece.northsouth.edu/people/dr-nabeel-mohammed